Set Sail for the Grand Line: Charting Android Apps with Agents and Android CLI



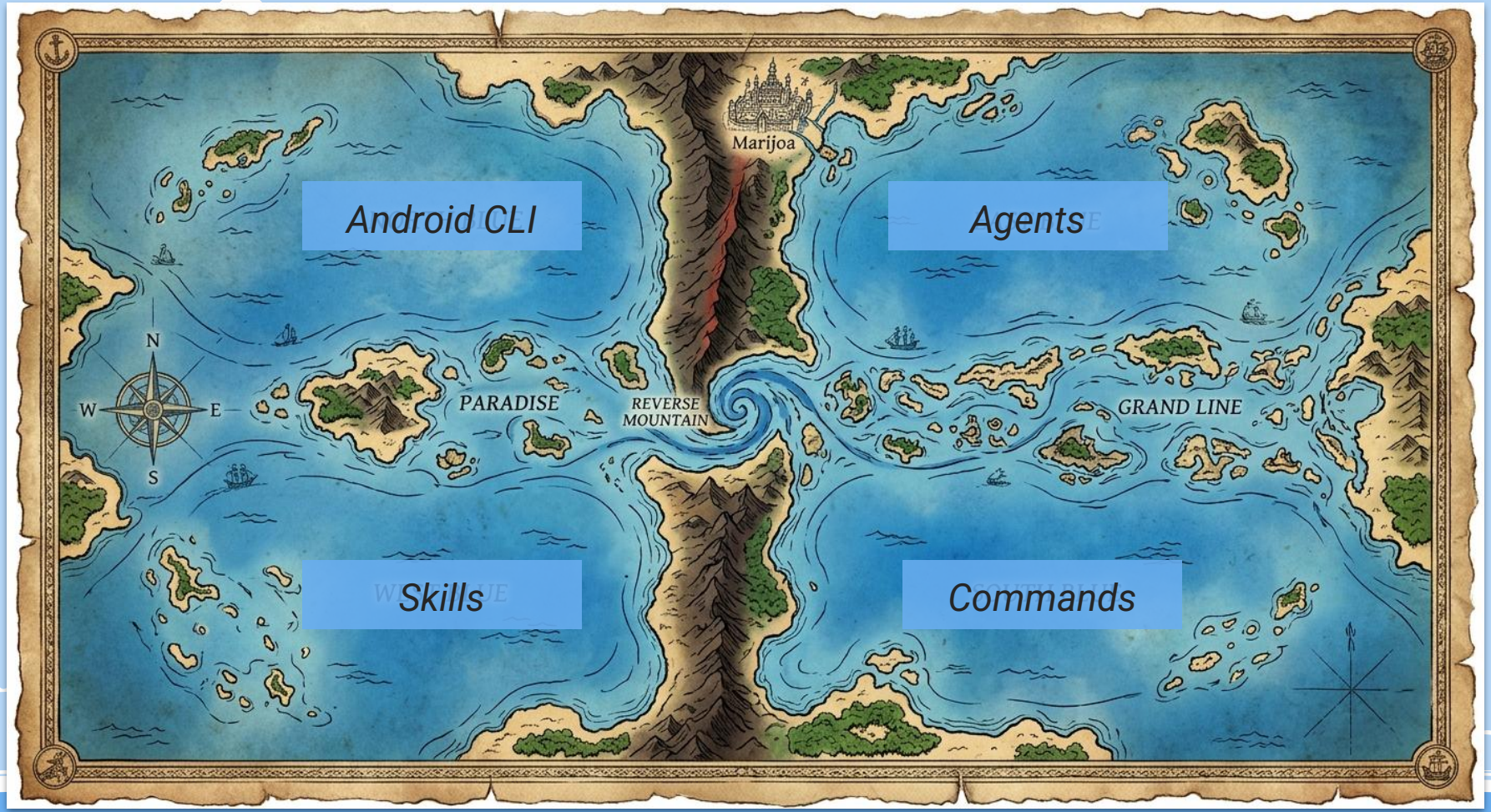


Ahoy!

- Android pirate since 2009
- GDE for Android since 2023
- Part of the crew in Medtronic

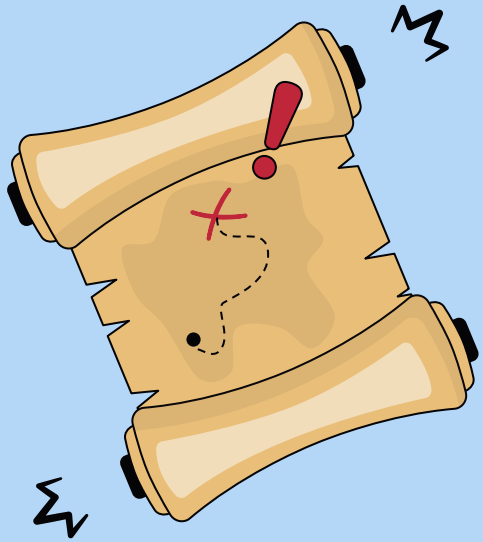
Preparación

https://tinyurl.com/ym3mh3pf


Android CLI
Agents
Skills
Commands
Marijoa
PARADISE
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GRAND LINE
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Android app





Let’s start our adventure!

Have you… ?



So what's the problem?

API Hallucinations

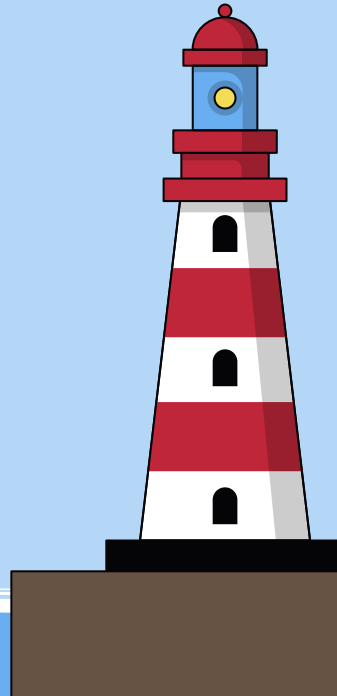
Agents frequently suggest deprecated Gradle options or outdated dependencies.

Context waste

Dumping entire .gradle.kts files or Logcat dumps burns thousands of tokens needlessly.

Fragmented tooling

Commands scattered across adb, sdkmanager, avdmanager, and custom Gradle tasks.



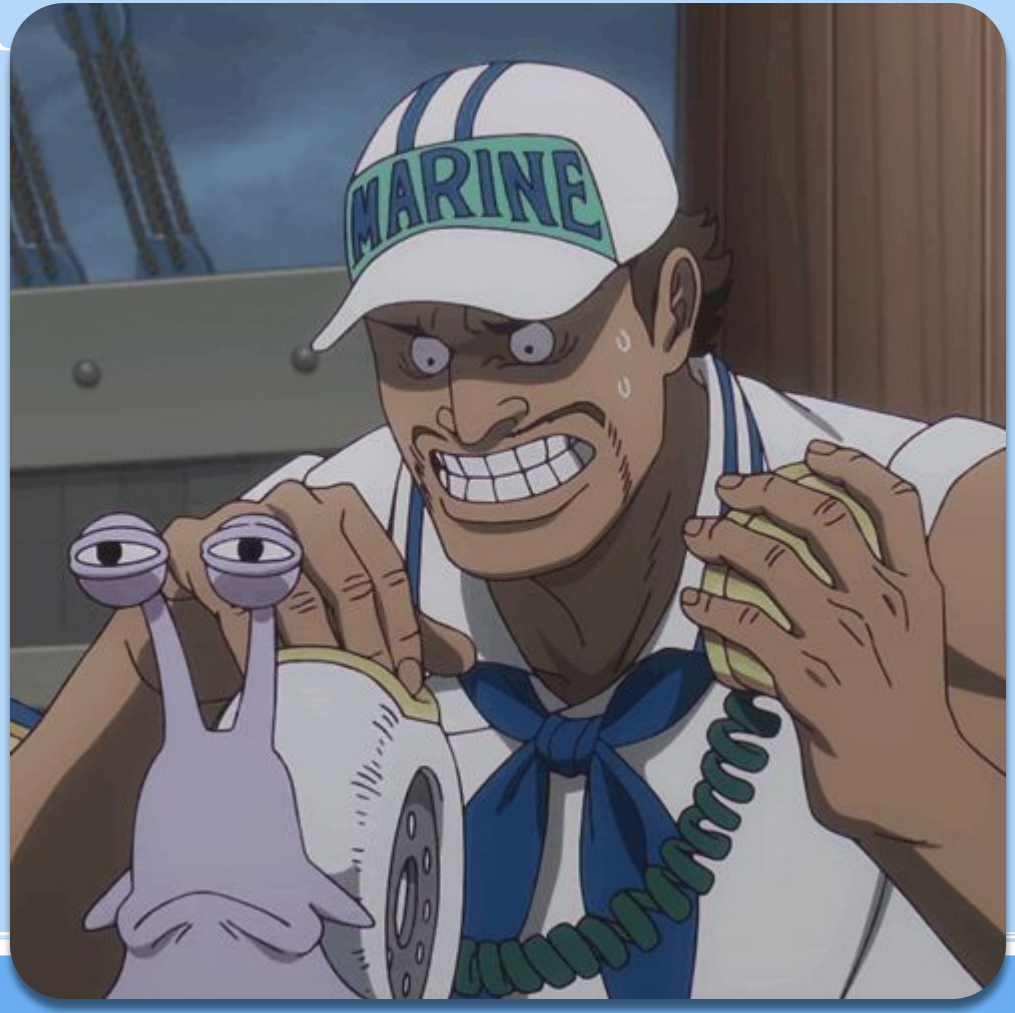
"Android Studio is a fantastic IDE for humans, but heavy and inefficient as a direct interface for AI agents."

Monkey D. Gemini

01 What is Android CLI?

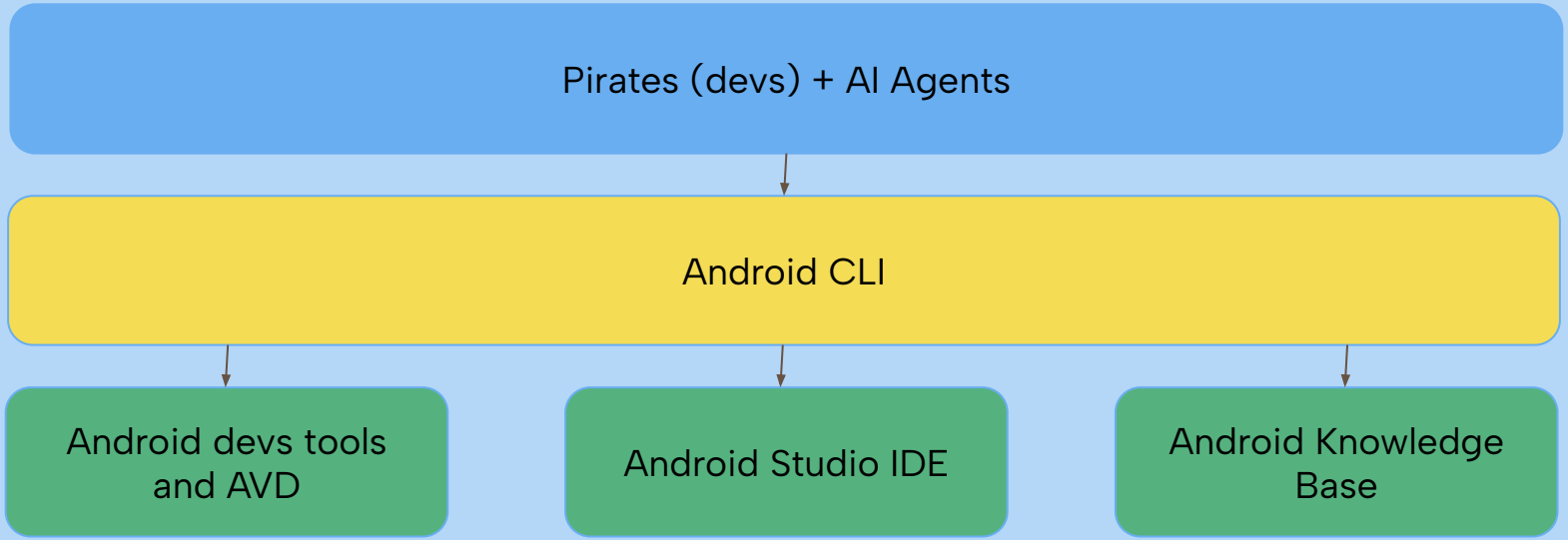


And how does it solve my problems?


MARINE

Android CLI

Android CLI serves as the standardized abstraction layer **connecting** developers and AI agents directly with the Android ecosystem.



Android CLI: Core pillars

Unified Commands

Replace complex multi-step workflows with **direct CLI commands** (android create, android run, android sdk).

Android Skills

Modular, executable knowledge packages codifying **official best practices** (Navigation 3, Compose, Edge-to-Edge).

Android Studio Bridge

Communicate with running IDE instances for AST analysis without rendering heavy UI components.

5 functional blocks

Environment & SDK Setup

Initializing, diagnosing, and maintaining the Android development environment

Project Management & Execution

Creating projects, inspecting structure, building, and deploying applications

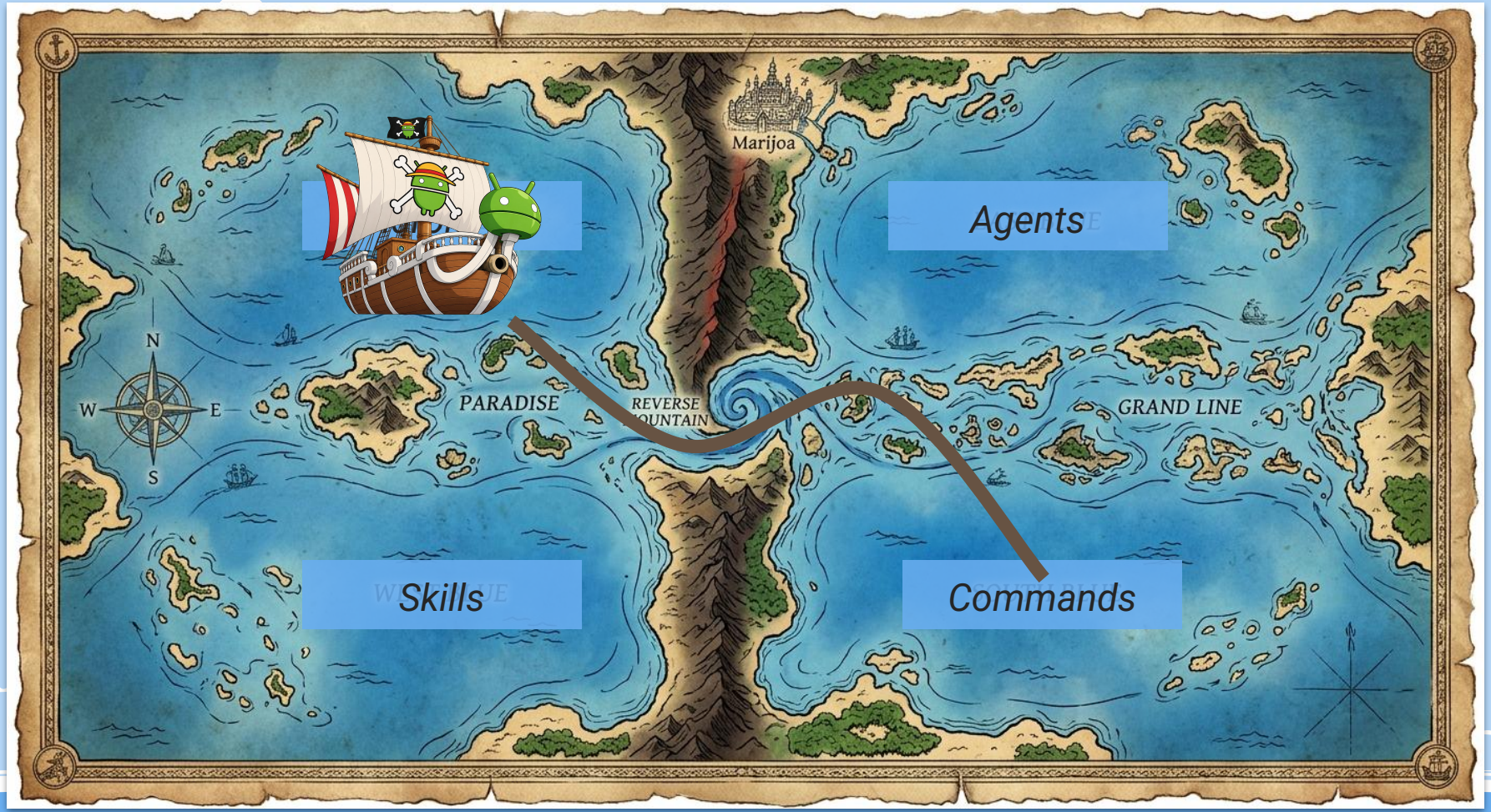
Devices & Emulators

Managing Android Virtual Devices (AVDs) and physical test hardware

UI & Studio integration

Analyzing component hierarchies, Compose views, and screen activity

AI Integration


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Commands

Let’s communicate!

Environment and SDK setup

android sdk
Installs, updates, removes and lists SDK packages.

android init
Sets up the Android CLI and installs the android-cli skill into the agents detected on your machine.

android info / update
Shows the environment details (SDK location, etc.) and updates the CLI to the latest version.



Project and management execution



android create
Bootstraps new Android projects using official templates.

android describe
Generates metadata detailing project modules and build artifacts.

android run
Deploys and launches an already-built APK on a target device

android install
Fast incremental APK installation without launching components.

Devices and emulators



android emulator
Creates, lists, starts, stops, and configures AVDs

android device
Controls connected physical and remote devices and queries status.

UI & Studio integration

android layout
Retrieves and inspects the active UI view hierarchy and Jetpack Compose component tree.

android screen
Captures screenshots and visually evaluates bounding coordinates.

android studio
Connects CLI directly to a running Android Studio instance to analyze code and render Compose previews



AI integration & documentation

android docs
Searches and retrieves official developer documentation and API guides directly from the terminal.

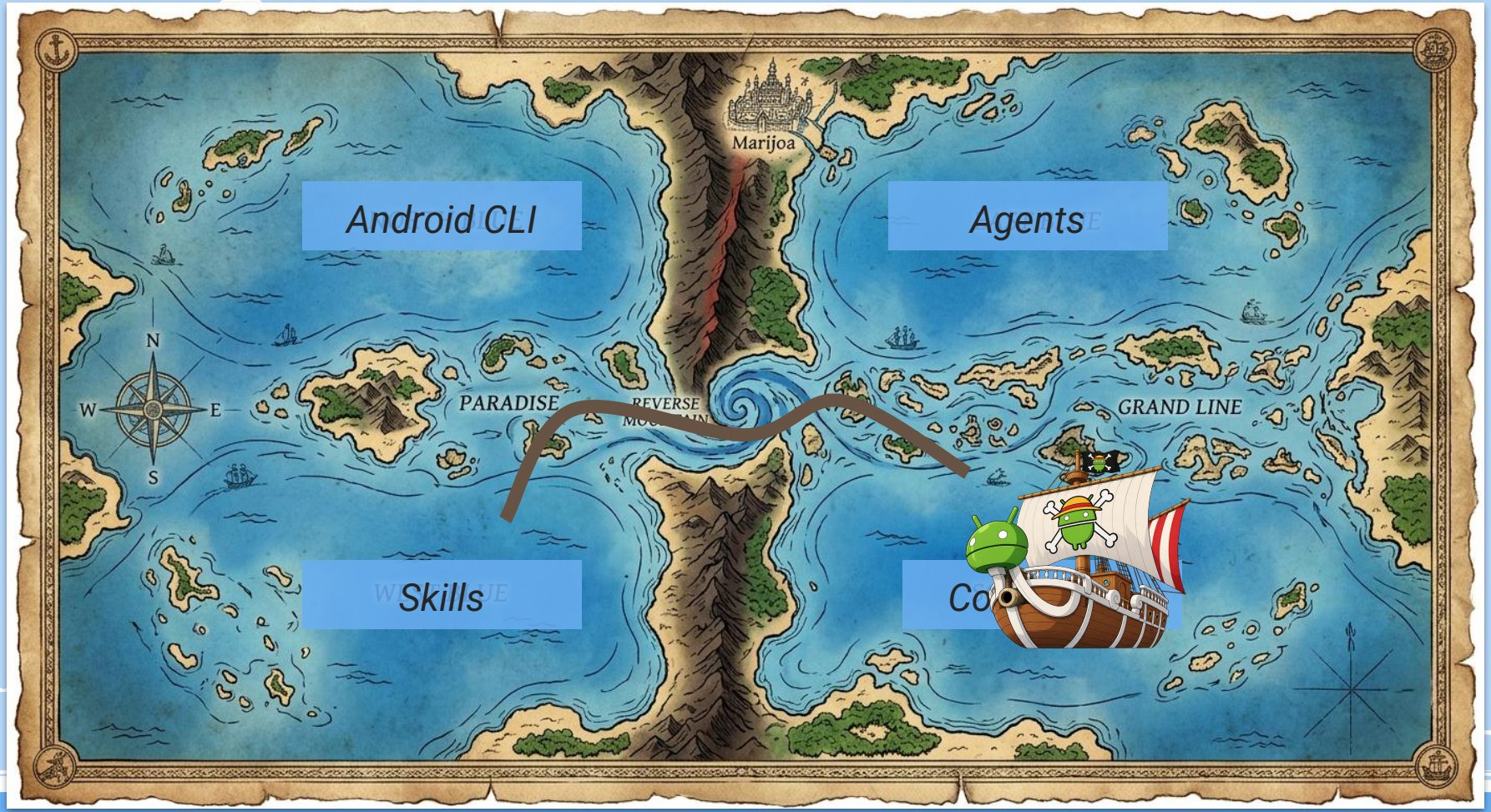
android skills
Manages agent skills and capabilities for automated AI development workflows.





Let’s see it in action

https://developer.android.com/tools/agents/android-cli


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03

Skills

Powers for our Agents



A skill is a self-contained, domain-specific instruction package that equips an AI agent with specialized knowledge, tools, or workflows to perform targeted tasks.

Portgas D. Claude



Android skills

Curated instruction packages that teach AI coding agents how to build Android apps according to Google's official best practices.

Why it matters:

- Architecture Enforcement
- API Correctness
- Cross-Agent Compatible



Official Android Skills

DevTools

android-cli (Teaches agents how to interact with the Android CLI)

Architecture

architecture, navigation (State management, ViewModels, and Type-Safe navigation)

UI & Design

compose-m3, wear-compose-m3, tv-compose (Material 3 and specialized UI platforms)

Testing

testing (Automated UI and unit testing patterns)



Managing the skills via Android CLI

android skills list
Lists all available and installed skills

android skills add/remove <skill-name>
Installs a skill into the project or removes it

android skills update --all
Updates all project skills to latest



https://github.com/android/skills

Official skills repository

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04

Agents

Our digital crew

Why agents struggle with Android?

High token cost

Sending massive source code files or large stack traces drains context windows and inflates costs

Hallucinations & outdated APIs

Without real-time environment feedback, LLMs generate code for deprecated libraries or incorrect configurations

No execution feedback (until you do it)

Agents normally write code without knowing if it actually compiles, installs, or renders as expected

Android CLI: Empowering Agents with Real-Time Capabilities



Android CLI & Agents: BFF

Precision tools over raw code

Agents query structural metadata instead of parsing thousands of lines of raw code

Grounding via official docs

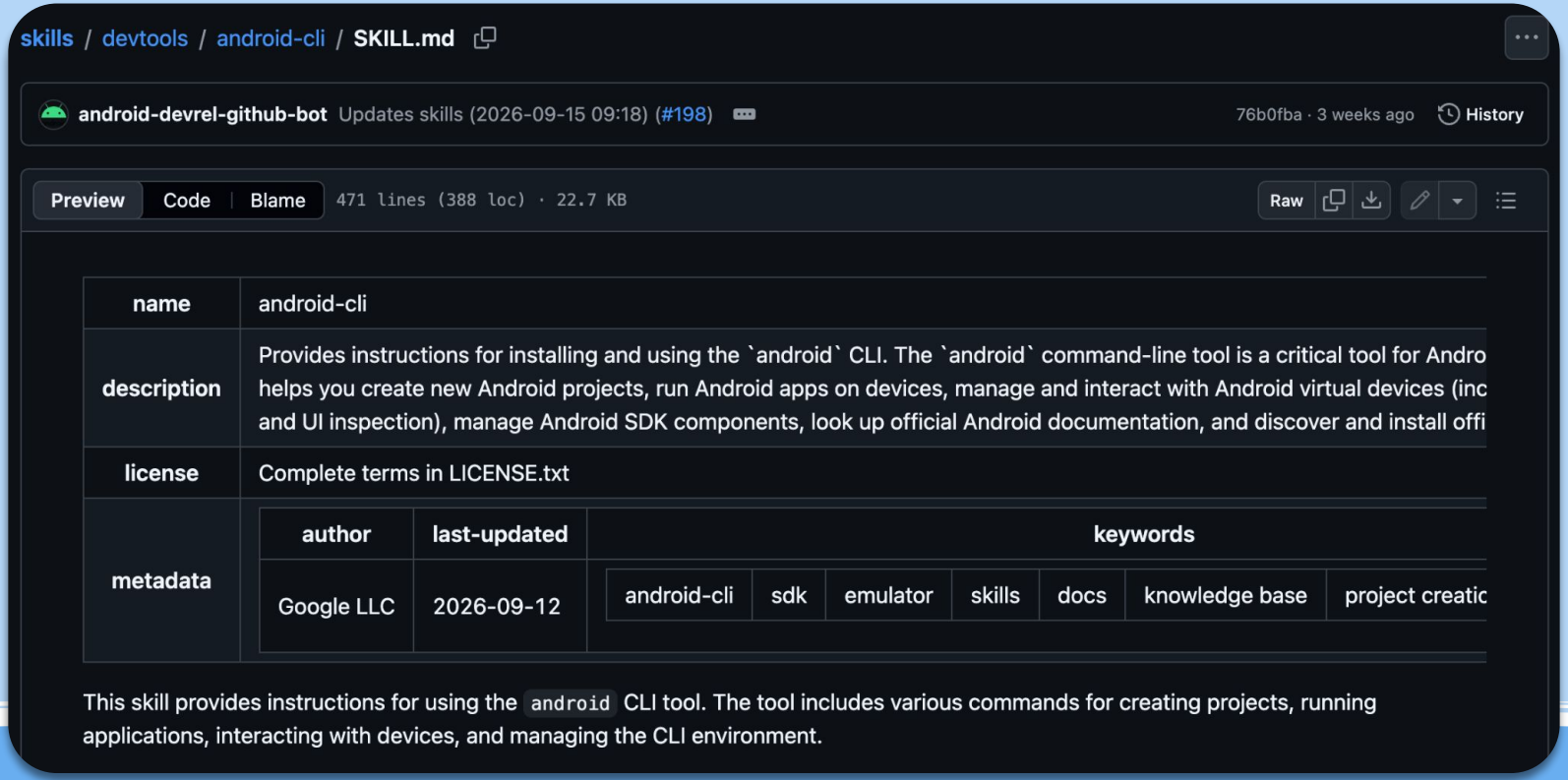
Direct access to updated docs grounds LLMs in current best practices

Standardized agent skills

Skill packages equip agents with pre-packaged domain knowledge (Jetpack Compose, MVVM, etc.)



The magic ~~fruit~~ skill



The best part

android init

Copilot cli

Antigravity cli

Claude code

What if I use another CLI? (codex?)

android skills add --all --agent=codex

Copy and paste the skills to .agents/skills/



Let’s see it in action

Sanji's Galley App

Starter:
https://github.com/gemasr/setsail_starter

Finished project in phases:
https://github.com/gemasr/sanjis_galley



The future of Android development is about giving AI the right tools to verify its own work. Android CLI makes agents faster, drastically reduces token costs, and lets them test their own UI before we even review the changes.

Gol D. ChatGPT


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With Android CLI and AI Agents as your crew, you are no longer just coding.
You will be charting the unknown seas!



Thanks!

Do you have any questions?

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